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Assignment Cover Page

BIO101

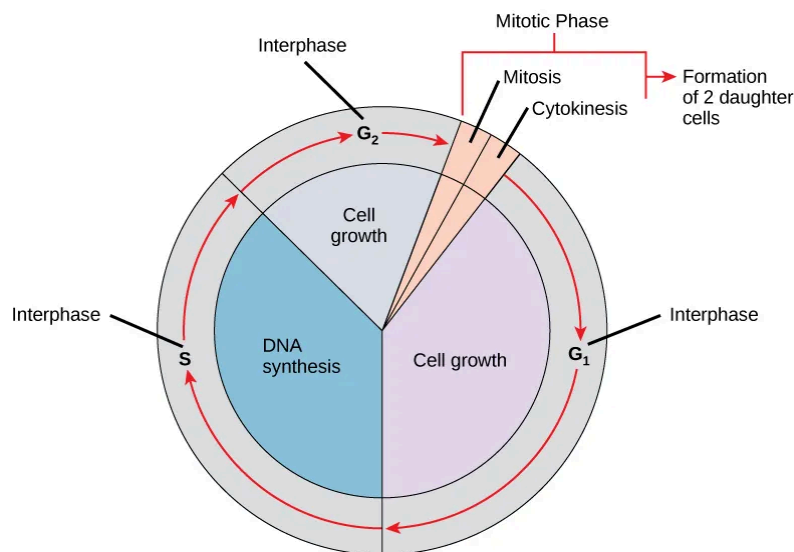
Introduction to Biology

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Course Title	Introduction to Biology
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1. Answer:

Growth, repair and replacement of damaged cells require cell division. According to our lecture slides, cell division is the process by which one parent cell divides to make two or more daughter cells; the cell cycle is an ordered series of growth and division events that leads to the formation of two new daughter cells. This is the typical process of a normal functional cell. But if the control system does not function properly, cells may start dividing when they shouldn't. An important characteristic of cancer is the unchecked growth of abnormal cells.

The cell division can be split into two phases: interphase and the mitotic phase (M phase). Interphase includes three sub-phases, G₁ (first gap), synthesis (S) and G₂ (second gap). In G₁ the cell grows and replicates DNA, and in G₂ the cell continues to grow and activate the molecular machinery required for cell division (Alberts et al., 2022). The M phase then carries out the actual division of the cell, by mitosis and cytokinesis.



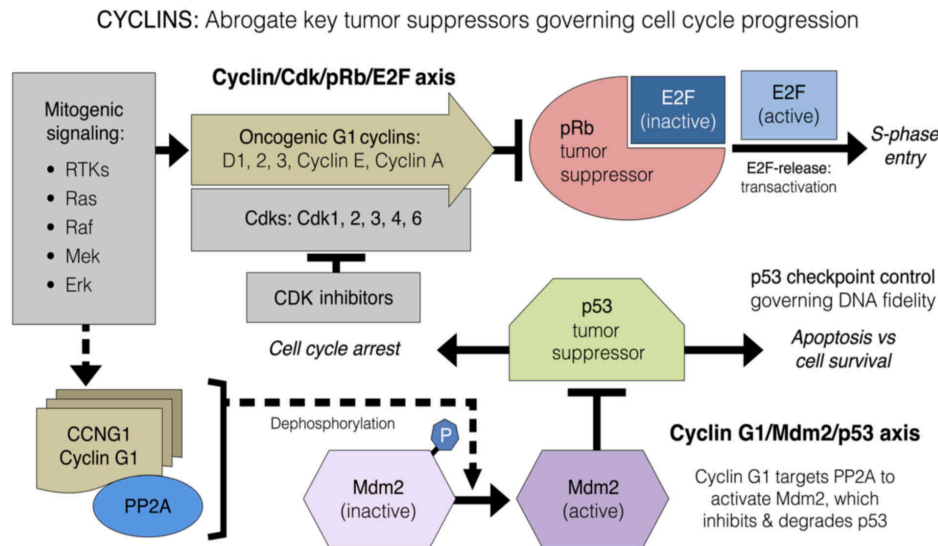
During the G₁ phase, the cell grows and determines whether the conditions are right to divide. The cell is required to be large, provide sufficient nutrients and transmit correct growth signals. DNA replication takes place in the S phase. The copying of DNA prior to cell division, so that the daughter cells receive an equal amount of DNA to the parent cell is explained in our Lecture

Slide. During the G₂ phase, the cell verifies DNA replication has occurred properly prior to starting mitosis.

The checkpoints are much like 'security gates'. They block the passage of damaged or abnormal cells from passing on to the next cycle of mitosis. If the DNA is damaged, then the cell can stop the cycle of division to correct the damage. When the damage is so extensive, the cell may commit programmed cell death (apoptosis). This prevents the damaged cells from reproducing.

There are three large checkpoints: the G₁/S checkpoint (also known as restriction point), the G₂/M checkpoint, and the spindle assembly checkpoint (SAC) in M phase (Weinberg,

2014). G1/S and G2/M checkpoints are particular points in interphase that warrant special attention to quality control.



Proteins serving as "switches" regulate the checkpoints. There are two important types: cyclins and CDKs (cyclin-dependent kinases). A cyclin can bind to a CDK, thereby activating it which in turn releases the "go ahead" signal for the cell to enter the

next stage. Each cyclin–CDK complex regulates a specific transition in the cycle (Morgan, 2007).

p53 and Rb are two important stopping signs. When the DNA in the cell is damaged, p53 is activated and stops the cycle, allowing the cell to repair the damage or, if it's too serious, causing the cell to commit suicide. Rb also functions in a similar manner, blocking genes that would direct the cell to copy the DNA. Once healthy and ready, Rb gets turned off and the cell continues to grow (Alberts et al., 2022).

The checkpoints fail to function if the proteins are damaged or lost because of gene mutations. The cell is no longer capable of recognizing problems and continues to divide, even if it shouldn't. That's what's at the heart of cancer.

Mutations in p53: Over half of all human cancers have mutations in p53 (Weinberg, 2014). Without it, the cells with damaged DNA continue to proliferate and transmit the errors to new cells over time, causing cancer.

Without Rb: The signal to begin copying DNA is always switched on. Cells do not cease dividing until an appropriate "go signal" is given. It was one of the first mutations known to cause cancers, and is associated with a childhood eye cancer called retinoblastoma (Weinberg, 2014).

If the cell makes too much of the cyclin protein, the "go" signal is always on and the cell races through the checkpoints. This is observed in various cancers, such as breast cancer (Alberts et al., 2022).

Cancer doesn't really happen via single mutation. In general, multiple checkpoint regulators need to be defective in order for a cell to become completely cancerous. With each failure, the cell acquires a small growth advantage and, with time, the abnormal cells outcompete the normal ones. This leads to a mass of cells that proliferate without any control and without caring about normal signals to stop growing, and the mass may metastasize to other parts of the body (Hanahan & Weinberg, 2011).

One such disease is colon cancer, which takes years to develop. One gene mutates, another mutates, and more and more, each successive mutation, at least in part, to be caused by earlier checkpoint failures and accumulated errors. The importance of these checkpoints is illustrated by the fact that losing any one of them is like losing the whole safety net and can begin the chain of events leading to cancer (Weinberg, 2014).

If key proteins, such as p53, Rb and cyclins are disrupted, cells lose this control, accumulate damage, and may develop into cancer. That is why it is crucial to understand the cell cycle in order to gain insight into and, hopefully someday, improve the treatment of the disease.

2. (a)

Answers:

The debate concerning animals in research for human purposes is one of the most controversial issues in bioethics, because it has helped medicine, but it also raises serious ethical concerns about animal suffering. The utilitarian argument that animal research can contribute to medicine and human suffering should not be ignored, while the deontological and rights-based perspective should be respected, that animals should not be used solely as tools. The use of animals for research, can only be done if provided the following conditions are met: it must be truly necessary; it must be well supported, carefully regulated; it must be done by a commitment to not use animals when alternatives are available, the best case being trying to find alternatives. This claim is endorsed by Festing and Wilkinson (2007) who stated that animal experiments can be justified if needed and when good ethical controls are adhered to.

The mouse genome shares about 85% of its protein-coding genes with humans and nearly all of the essential biological pathways — such as regulation of cell cycle, immune regulation, hormone signaling and metabolism — are going by similar mechanisms at the molecular level (Shanks et al., 2009). This translates into work for some in mice, being biologically relevant at the molecular level in humans, at least.

Inbred strains of mice or rats allow for genetic uniformity and help researchers reduce various differences between individuals and allow for the isolation of the effect of one experimental issue. Their relatively short generation time (20 days for mice) also permits multigenerational studies within practical time frames (National Research Council, 2011).

Although useful, cell cultures cannot replicate the whole organism; pharmacokinetics, immune response, organ level toxicity and interactions between multiple systems in the body are excluded. The major difference between cell-based models and animals is that animals offer an integrated biological system that a cell-based models cannot (Shanks et al., 2009).



The discovery and development of insulin is one of the most important event for medicine to come as a result of animal research. In 1921, Frederick Banting and Charles Best at the University of Toronto were able to show, using dogs, that extracts of the pancreas were able to cause the blood glucose to decrease in diabetic animals. They then refined this extract with the help of cattle pancreases, and were able to test this purified extract in a diabetic dog before giving it to a human patient, fourteen-year-old Leonard Thompson, in January 1922 (Bliss, 1982). The treatment was the making of his life and indeed turned Type 1 diabetes from a life-threatening condition to a healthy level one.

This example shows how animal testing made an important contribution, because

the dog model provided early proof that insulin could work in a living body. Safety data, and dose information that was not available from any animal model at the time. It was the discovery that eventually won Banting and Macleod the 1923 Nobel Prize in Physiology or Medicine. The modern recombinant human insulin, made in bacteria, was also tested in animal models before it was widely used in humans (National Research Council, 2011).

The 3Rs Framework

The scientific community acknowledges the ethical responsibility for the use of animals, and follows the "Three Rs" principle of Russell and Burch (1959): Replace non-animal alternatives, Reduce the number of animals used in a study through careful experimental design, Refinement of procedures to minimize pain, suffering and distress. This framework is now incorporated into many parts of the world, including Europe, North America, and much of Asia, as part of the regulations (National Research Council, 2011).

2. Answer:

In a general sense, the sum total of the contribution of animal research to human (and animal) health is huge. Animal models were used to develop, or validate, vaccines for polio, rabies, etc., surgical methods, anaesthetics, cancer chemotherapies and HIV antiretrovirals (Paul et al. 2010). The suffering prevented and lives saved in many populations of humans and domestic animals is a moral event.

Most importantly, at the moment, there is no full in vitro or computational alternative to the complexity of a living mammalian system for safety. This is why regulatory agencies, like the U.S. Food and Drug Administration (FDA) and the European Medicines Agency (EMA), require pre-clinical animal studies before granting approval for human clinical trials. That is why whole-organism models are required to assess systemic toxicity, pharmacokinetics, and immune interactions before the approval of drugs for human clinical trials by regulatory agencies such as the FDA and EMA. The only way to eliminate animal testing would be to do so safely, and under current scientific abilities, this would result in the lack of life-saving drugs — with its own moral obligations.

There are solid philosophical arguments against animal testing. Peter Singer's utilitarian argument suggests that if suffering is morally significant for humans then it would be for sentient animals, that is, animals of any species that can feel pain or distress; the species of the sufferer is not enough by itself to render him or her morally significant (Singer, 2009). In extending the rights-based argument, Tom Regan asks what is the value of an animal if it is a 'subject of a life', that is, if it has beliefs, desires, perception, and a welfare; what is the value of that if it cannot be exchanged for human good? (Regan, 2004).

There have been times in the history of animal research where unnecessary suffering, multiple research repeated, and poor animal welfare have occurred. Cosmetic testing on animals is a case in point where the benefit (cosmetics) is clearly not worth the harm (animals), and there is an ethical consensus against it, which is rightfully progressing against it. We should not allow this and stop according to our capabilities and not support them in any way.

So, the ethical acceptability of animal research is not a matter of 'right versus wrong', but rather depends on the context. It is acceptable where:

- (1) The human or animal health benefit is significant;
- (2) There are no suitable non-animal alternatives to its use in the given question;
- (3) All of the 3Rs are strictly adhered to to minimise the number of animals used and the amount of suffering they endure; and
- (4) The research is independently ethically reviewed. It doesn't make sense when cosmetics are being used or if there are alternatives that have been validated and/or welfare standards are low.

Finally, animal testing is a pivotal point for biomedical research but definitely a tough issue to be solved as it has a moral perspective as well. Rather than stopping completely, the answer to this issue is ongoing ethical questioning, funding of alternative technologies (organoids, microphysiological systems, and AI-based prediction of toxicity), and consistent implementation of the 3Rs, and ultimately, the complete elimination of animals as alternatives improve.

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